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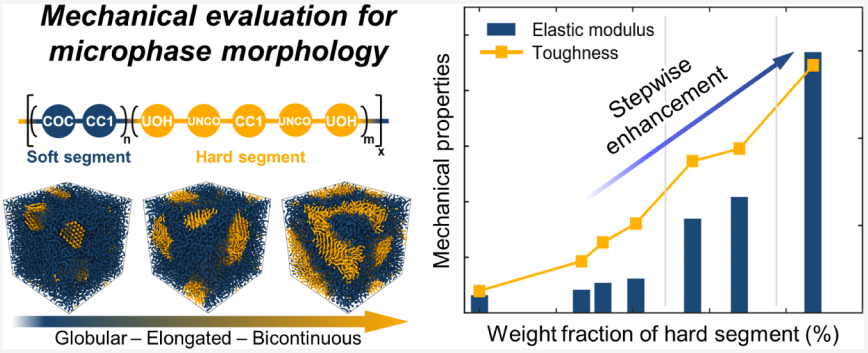
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Molecular-Scale Investigation of the Microphase-Dependent Load Transfer Capability of Polyurethane

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ABSTRACT: In this study, we investigated the mechanism by which the microphase structure of polyurethane (PU), manipulated by the chemical composition, determines its macroscopic mechanical properties. Increasing the hard segment content induced a microphase transition from globular to elongated to bicontinuous. This transition significantly altered the mechanical behavior of PU from hyperelastic to elasto-plastic. This enhancement in the mechanical properties was related to the load-transfer capacity of the hard domains in each microphase. In the globular phase, most of the strain energy was absorbed by the soft matrix, limiting the contribution of the hard phase to the mechanical properties. Conversely, elongated discontinuous structures facilitated a homogeneous strain distribution during tension, promoting an immediate load transfer to the hard domain. To quantitatively evaluate the load-transfer efficiency, a mechanical model in which one soft hyperelastic spring was coupled to two rigid elasto-plastic springs was considered. The effects of the microphase morphology and hard domain dissociation on the load-transfer capability were identified. This study contributes to a molecular-level understanding of the deformation behavior and mechanical response of microphase-separated PU.

1. INTRODUCTION

Polyurethane (PU) is a block copolymer composed of alternating hard and soft segments.[1−3] Due to the thermodynamic incompatibility between hard and soft segments, each segment spontaneously separates into individual phases,[4−6] forming a nanodomain with a spatial scale of ∼20 nm.[7−9] The hard segments are strongly engaged with each other by hydrogen bonds between urethane groups and exhibit higher structural rigidity than the soft segments.[10,11] Therefore, the hard domains act as homogeneous nanosized reinforcements in the soft matrix under mechanical loading, providing PU with mechanical strength and toughness.

One characteristic that arises from these structural features is that the mechanical properties of PU vary significantly depending on the microphase-separated morphology.[12−15] In particular, the mechanical responses of PU show nonlinear transitions depending on the chemical composition, such as hard segment content or the soft chain length. Tien and Wei synthesized PU polymers with hard segment contents of 39−63 wt % and evaluated their tensile properties.[16] When the hard segment content increased from 39 to 50 wt %, the material showed a steep increase in elastic modulus (218 to 1324 kgf/cm^2). Further increase in content led to additional stiffening, but the reinforcement effect was considerably diluted (1679 kgf/cm^2 in 63 wt %). Similar observations were also made in a study by Chen et al., who fabricated PU with different hard segment weight contents from 31.5 to 55.0 wt %.[17] At low hard segment concentration, the tensile modulus of PU was comparable to the soft skin (3.6−7.9 MPa in 31.5−45.7 wt %). However, as the hard segment content exceeds a threshold point, the overall elastic modulus improves very rapidly (30.4−42.1 MPa in 50.7−55.0 wt %). Zeng et al.

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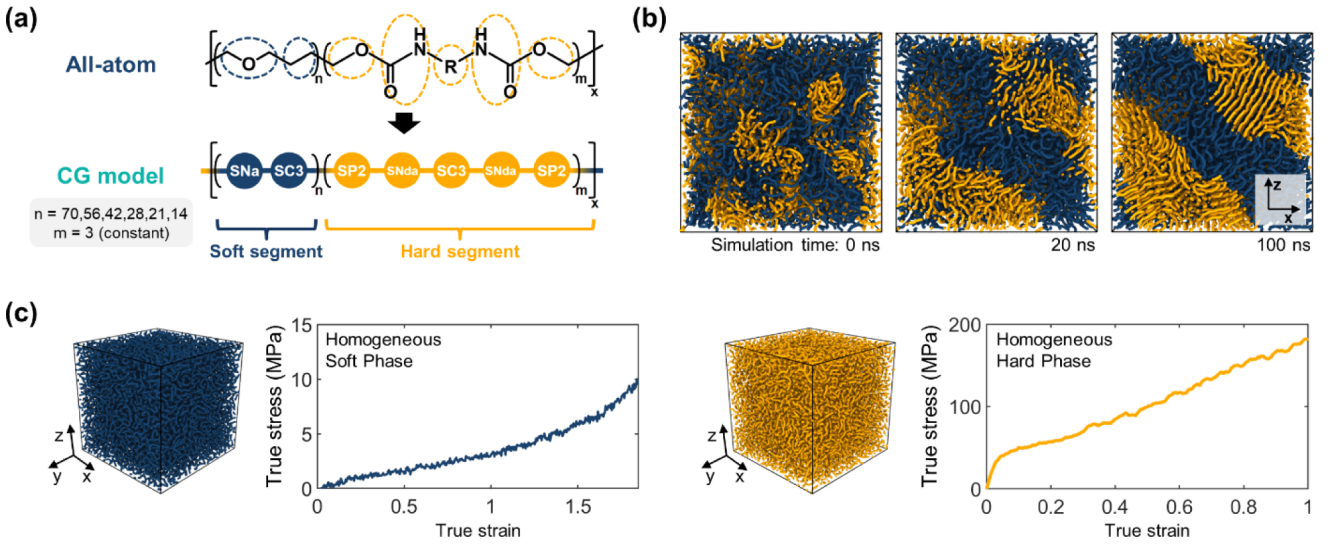


Figure 1. Preparation of microphase-separated PU structures. (a) All-atomic chemical structure and MARTINI-based coarse-grained mapping of the PU molecule. R is the backbone ring of the isophorone diamine. (b) Molecular configuration change during phase separation simulation. (c) Evaluation of mechanical responses of homogeneous soft and hard phases under uniaxial tensile deformation.

similarly observed nonlinear improvements in tensile strength and fracture toughness of PU under analogous conditions.[18]

Considering the stiff and glassy nature of the hard segment, such a dramatic transition is closely related to the change in the load transfer capability of the hard domain for the microphase. It was found that the phase separation structure, which changes depending on composition conditions, plays an important role in the mechanical behavior.[19−21] The lateral growth of hard domains and interconnected morphology contribute to the mechanical improvement of PU. Recently, Huang et al. suggested that the nonlinear mechanical transition is associated with a morphological change from sea−island to cocontinuous structures.[22] Using tapping-mode atomic force microscopy, the microscopic Young's modulus distributions of phase-separated PU films were directly measured, inferring that continuously distributed hard segments contribute to the macroscopic modulus improvement. Nevertheless, due to the limited spatial resolution of experimental techniques, clearer evidence at the molecular level is needed to further support the mechanism.

From soft, rubber-like to rigid and stiff, a wide range of properties make PU suitable for a variety of engineering applications, including pressure-sensitive adhesives, sealants or coatings, biomedical scaffolds, and noise or vibration damping components.[23−28] Therefore, to further expand the engineering potential, it is essential to precisely manipulate the properties by exquisitely designing the microphase structure. However, the correlation between the microphase structure and macroscopic mechanical properties remains elusive. In particular, understanding the microphase-dependent load transfer mechanisms in hard domains is of utmost importance.

Motivated by the above, we investigated the load-carrying capacity of hard segments depending on the microphase morphology. Microphase-separated PU structures were fabricated with various hard segment contents in a coarse-grained molecular dynamics environment. The implemented PU models exhibited a stepwise phase evolution from a globular to a bicontinuous structure with increasing content. Importantly, the mechanical properties were also greatly affected by microphase evolution. For an in-depth assessment of the quantitative contributions of the individual phases, subcontinuum analysis and mechanical modeling were employed. Therefore, the present study contributes to elucidating the microstructure−property relationships for rational PU design.

2. MODELS AND METHODS

2.1. Molecular Modeling of Microphase-Separated PU. Coarse-grain modeling of the PU molecules was performed using commercial MD software Materials Studio 2020 (Dassault Systèmes, France). Phase-separation simulations and mechanical evaluations were performed using the open-source code LAMMPS (Sandia Lab).[29]

The chemical composition of PU was adopted from the literature.[30] The hard segments of PU were prepared from three isophore diisocyanates linearly extended with ethylene glycol (Figure 1a). Poly(tetramethylene ether glycol) was used as the soft segment interconnecting the hard segments. To vary the hard segment content within the molecule, the length of the soft segments was varied in the range of 14−70 repeat units; thus, the hard segment content ranged from 13.3−43.5 wt %. The total molecular weight was set to approximately 30 000 g/mol, regardless of the composition, such that all PU molecules exhibited similar characteristic lengths. Therefore, the influence of the chain length on the microphase and mechanical properties was completely excluded in this study.

Coarse-grained mapping of the PU molecules was conducted using the MARTINI force field.[31,32] The bead mapping of the PU molecules in this study followed the MARTINI standard coarse-graining.[33−35] The number of atoms required for coarse-graining into a single bead is not constant, and the number of heavy atoms participating varies from 2, 3, or 4. In particular, the assignment of an atom bundle to a bead is made based on the polarity of a specific group and its ability to form hydrogen bonds.

The MARTINI force field provides a simple, consistent parameter for covalent bonds consisting of the polymer backbones. In other words, the covalent bond term between two adjacent beads and the angular term between the three beads are considered in the potential energy. For noncovalent interactions, the MARTINI force field categorizes bundles of atoms into four types: polar (P), nonpolar (N), apolar (C), and charged (Q). These bead types are further subdivided based on their hydrogen-bonding capabilities (d = donor, a = acceptor, da = both, and 0 = none) and degree of polarity (assigned as an integer from 1 to 5, depending on the magnitude).[32] Also, the MARTINI force field uses a Lennard-Jones 12−6 potential energy to describe the noncovalent interactions. While the distance at which the

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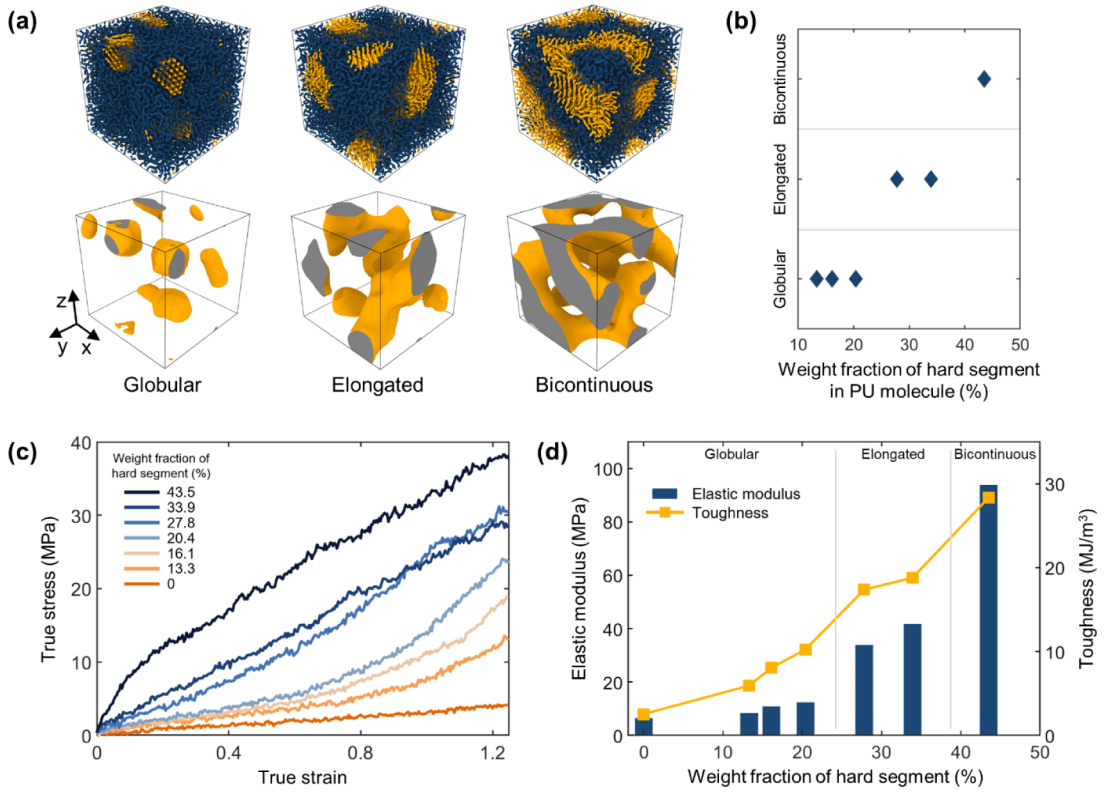


Figure 2. Evaluation of tensile mechanical properties of PU according to the microphase structure. (a) Configurations of equilibrated PU microstructures (top) and highlights of hard phase distribution (bottom). In the configuration, dark blue and yellow beads represent the soft and hard segments, respectively. The hard segment content is 20.4, 33.9, and 43.5 wt % from the left. (b) Microphase evolution as a function of the hard segment contents. (c) Stress–strain profile of PU derived during the uniaxial tensile test and (d) changes in elastic modulus and toughness.

particle–particle energy becomes zero is constant at 0.47 nm, the depth of the potential well is divided into 10 levels. Typing of the noncovalent parameters at the coarse-grained level is also based on the MARTINI standard coarse-graining.[33] The detailed noncovalent parameters used are summarized in Tables S1 and S2. Such approaches based on MARTINI force fields are known to provide reliable results for the separation and reorganization of polymer microphases.[36] Hyperelastic behavior observed in soft materials such as elastomers has also been successfully reproduced and is discussed in detail in Section 2.2.

The 20 PU molecules prepared with different hard segment contents are randomly positioned in a cubic simulation cell. To avoid unintended interference between molecules during initial positioning, unit cells are prepared at a very low density of 0.1 g/cm^3. An NVT ensemble at 300 K is performed using the Verlet algorithm with a time step of 5 fs for 5 ns. Subsequently, the models are progressively densified over 10 ns in the NPT ensemble at 300 K and 1 atm. After sufficient condensation, the models are annealed at a high temperature of 500 K for 10 ns. In this process, the individual phases that make up the unit cell (soft and hard phases) are completely mixed to form a homogeneous amorphous structure. Afterward, the model undergoes a long relaxation of 100 ns in an environment of 300 K and 1 atm, and the microphases of the PU structure gradually separate (Figure 1b).

2.2. Evaluation of the Effective Mechanical Properties of Hard and Soft Segments. Verification of the applied coarse-grained parameters was performed based on the effective mechanical properties of the individual segments. A molecular chain composed of only hard or soft segments was prepared, and a homogeneous-phase microstructure was fabricated according to the procedure described in Section 2.1. After sufficient relaxation, the effective mechanical responses of the hard and soft segments were evaluated by applying uniaxial tensile strain to the equilibrated model. While the models were deformed at a constant true strain rate of 10^8/s, the stress tensor on the constituent atoms was derived using the virial theory as follows:[37]

$$\boldsymbol{\sigma} = \frac{1}{V}\left(-\sum_{i}^{N} m_i(v_i v_i^T) + \frac{1}{2}\sum_{i}^{N}\sum_{j\neq i}^{N} \boldsymbol{r}_{ij}\boldsymbol{F}_{ij}\right) \quad (1)$$

where V is the volume; N is the number of constituent atoms; m_i and v_i are the mass and velocity of the i-th atom, respectively; $\boldsymbol{r}_{ij}$ and $\boldsymbol{F}_{ij}$ indicate the relative position and interaction forces between the i-th and j-th atoms, respectively.

Completely different responses were observed depending on the segment (Figure 1c). The soft segments displayed hyperelastic behavior characterized by a steepening slope in the stress–strain profile with increasing strain. There was no clear elastic–plastic transition, and the stress levels remained remarkably low, not exceeding 10 MPa. The elastic modulus of the homogeneous soft phase obtained by linearly fitting the profile up to a 2.5% strain was 3.56 MPa. This is very close to a typical estimate of the room temperature modulus of macrodiol of 3 MPa.[38] In contrast, the hard segments exhibited linear strain hardening with a distinct yield point. The stress level was 2 orders of magnitude higher than that in the soft phase. The elastic modulus of the homogeneous hard phase was found to be 1.18 GPa. Previous studies on the hard domain of urethane-containing polymers reported that the relevant tensile moduli reached 2.5–5 GPa,[39,40] similar levels to the current study. Consequently, these results closely align with prior estimates, emphasizing the contrasting mechanical roles of hard and soft phases:[41] Soft segments provide high deformability and elasticity to the material, whereas hard segments are the mainstays of mechanical stiffening.

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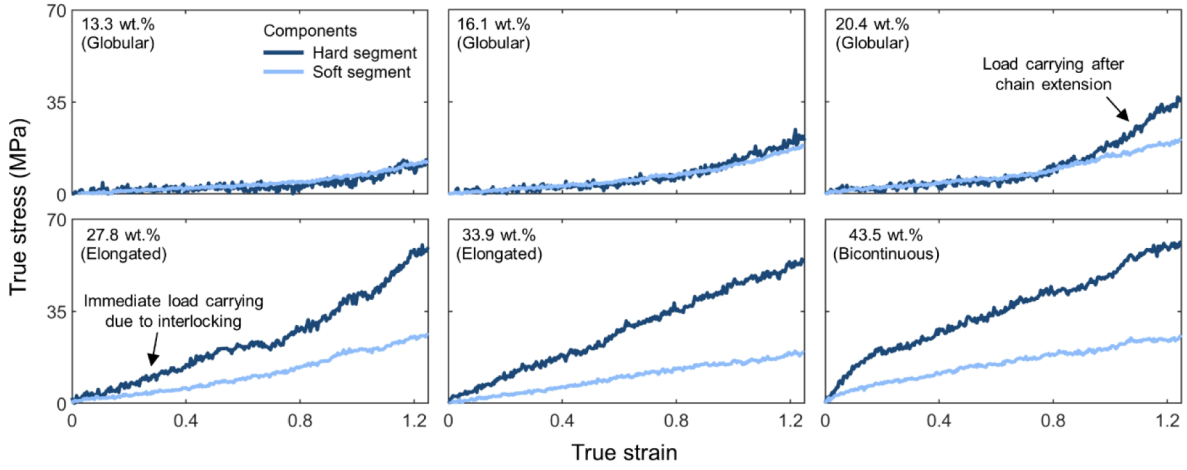


Figure 3. Atomic stresses generated in each hard and soft phase during uniaxial tensile deformation.

3. RESULTS AND DISCUSSION

3.1. Mechanical Evaluation of PU According to the Microphase Structure. During gradual phase separation, a variety of nanostructural morphologies were observed, ranging from globular to discrete elongated to bicontinuous phases (Figure 2a,b). When the hard segment content is low (13.3−20.4 wt %), the hard segments form spherical aggregations completely surrounded by a soft matrix. Increasing the hard segment content increases the diameter of the hard spheres, but their distribution remains separated (Figure S1). As the hard segment content within the PU molecule increases further (27.8−33.9 wt %), discrete globular nanodomains begin to coalesce, producing long and interpenetrating morphologies. Thus, introducing additional hard segments strengthens the connectivity between the hard domains. After exceeding the critical content threshold (43.5 wt %), the hard segments become infinitely connected across the periodic boundaries, leading to a bicontinuous morphology of soft and hard phases.

The influence of the microphase structure on the mechanical properties was analyzed. Uniaxial tensile tests were performed using the procedure described in Section 2.2. The microphase morphology of the hard segments significantly affected the overall mechanical response of PU (Figure 2c). When the hard domain is globular, the stress−strain behavior exhibits hyperelastic characteristics similar to those of the homogeneous soft phase. Notably, compared to the homogeneous soft phase (that is, 0% of the hard segment), the strengthening effect attributed to the hard segment is observed only at high strain levels (ε_T > 0.8). However, as the hard domains interconnect and elongate, mechanical behavior improves even at low strains. This is particularly evidenced by the fact that the elastic modulus and toughness of PU rapidly increase with phase changes (Figure 2d), becoming more pronounced as the PU microphase achieves bicontinuity. In addition, the stress−strain profile of PU undergoes a significant transition to elasto-plastic, similarly to that of homogeneous hard segments.

3.2. Estimation of Effective Contribution of Soft and Hard Segments. Overall, the mechanical properties of the PU materials exhibited a stepwise enhancement with evolution of the microphase morphology. The overall trend of the stress−strain profile also significantly alters from hyperelastic, similar to the homogeneous soft phase, to elastoplastic, similar to the hard one. In other words, the mechanical contribution of the individual segments, which strongly depends on the morphology of the microphase, dominates the macroscopic response of PU. These findings highlight the importance of considering the microphase structure in material design for effective mechanical modulation. Therefore, in this section, we individually evaluated the load-carrying capability of the PU phases based on their microstructures.

The effective load-carrying capabilities of the subdomains, the hard and soft phases, were evaluated using the subcontinuum methodology introduced in our previous studies.[42,43] The stress tensor, presented in eq 1, determines the load applied to the atom from the velocity obtained by thermal fluctuations and interatomic interaction forces. According to the virial stress definition, the interaction forces arising in each pair through covalent or noncovalent interactions are equally distributed to the participating atoms, irrespective of the atom type. Denoting V as the volume of the entire unit cell, it is self-evident that the stress exerted on the entire system is equal to the linear sum of the stresses allocated to each constituent atom. However, if V is reassigned to the Voronoi volume of its subdomain,[44] the stress experienced in the entire system is arithmetically equal to the volume-weighted average applied to each phase, as follows:[45−49]

$$\sigma_{Sys} = \sigma_H \Phi_H + \sigma_S \Phi_S \tag{2}$$

where σ is the virial stress and Φ is the volume fraction of the subdomains. The subscripts Sys, H, and S denote the overall system and hard and soft domains, respectively. This is because Voronoi tessellation defines the effective volume of a subdomain by halving the interatomic distance, regardless of the atom type, similar to the definition of virial stress. That is, the methodology allows the effective mechanical contributions of the hard and soft phases to be assessed separately in a single system regardless of its composition. Also, to highlight the asymmetric deformation between the hard and soft phases, a spatial strain distribution was additionally obtained. For a set of atoms 15 Å from the reference, the Green−Lagrangian strain tensor ($\boldsymbol{E}$) was calculated, as follows:

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) \tag{3}$$

where $\boldsymbol{C}$ and $\boldsymbol{I}$ indicate the Cauchy−Green deformation tensor and identify matrix, respectively. The derived strain tensor was scalarized to the von Mises strain.

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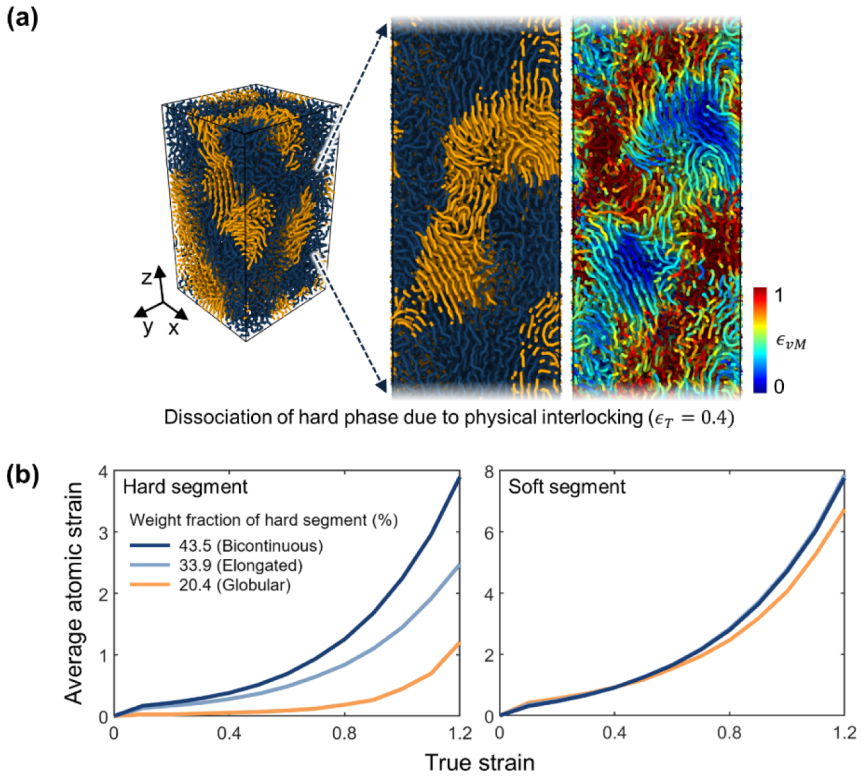


Figure 4. Atomic strain analysis during uniaxial tensile deformation. (a) Von Mises strain distribution at ε_T = 0.4 in the PU model with the bicontinuous hard phase (43.5 wt %). (b) Von Mises strain allocated to hard and soft segments according to their microphase structure.

The stresses developed in the hard and soft phases during tensile deformation were estimated separately (Figure 3). When the hard segment is aggregated into a globular structure, a high degree of similarity in the stress profiles between the soft and hard phases can be observed. Only at high strains of ε_T > 0.8, a distinct gap in the profiles is observed for the model with a higher hard segment content (20.4 wt %). This indicates that within the globular microphase, the external load is predominantly buffered by the soft domain. The hard domains experience substantial loading only after the PU chain is fully extended at an elevated strain. Conversely, when the hard domains are interconnected to form an elongated or bicontinuous morphology, a significant portion of the load is transferred to the hard segment in the early stages. This is because the irregular structure formed by the interconnected hard domains is favorable for load transfer. Therefore, in the bicontinuous microphase, an elasto-plastic behavior similar to that of homogeneous hard segments is observed in the hard segment profile.

Similar observations were made for strain distribution analysis. The Green−Lagrangian strain tensor captures variations in the relative distance between constituent atoms, excluding rigid-body displacements such as translation or rotation. Therefore, this approach highlights the inhomogeneous distribution of strains, depending on the segment type during the tensile process. Three main findings were obtained (Figure 4a,b). First, the soft segments exhibit a higher deformability than the hard segments during uniaxial tension. Such deformability characteristics are highly consistent regardless of the compositional conditions. In other words, the hyperelastic properties of the soft segment are largely preserved, irrespective of the microphase structure of the hard domain. This result is reasonable considering that the soft segments preserve a bicontinuous structure that is sufficiently interconnected, regardless of the content of the hard domains. Second, the hard phase with a globular structure experiences minimal internal deformation during tension. This is largely consistent with previous observations that the external loads are mostly buffered by the surrounding soft matrix. The strain of the hard phase becomes significant only after the PU chain has sufficiently expanded at ε_T > 0.8. Third, the interlocked structure, that is, elongated and bicontinuous, renders the strain distribution between the hard and soft phases more symmetrical. Therefore, the distortion and dissociation of the hard domains are easily observed, even at low strain levels.

3.3. Quantification of Load-Transfer Capability of the Hard Domain. The above results clearly demonstrate that the mechanical participation of hard segments varies significantly depending on their morphology within the PU. When the content of the hard segment is low, most of the external deformation is preemptively absorbed by the soft domain and the hard segments simply share the same stress magnitude as the soft ones. Such mechanical behavior can be better understood by considering the soft and hard segments as individual nonlinear springs, each with its own characteristics (see Figures S2 and S3). However, as the content of hard segments increases, the interconnected structure of the hard segment results in a more immediate response to external deformation. The global deformation imposed effectively deforms both domains; therefore, the PU behaves as a series of springs connected in parallel.

Inspired by these observations, we further performed mechanical modeling of the heterogeneous structure of PU. The point is that the load-transfer capability of the hard phase, which governs the macroscopic mechanical behavior of PU, is sensitive to both the microscopic structure and applied strain.

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In this regard, we considered a mechanical model in which one soft hyperelastic spring was coupled with two hard elasto-plastic springs (left side of Figure 5a). Two hard springs were arranged in parallel or series with one soft spring, each weighed by a variable volume fraction (Φ). In such a combination arrangement, each hard spring performs a separate mechanical role: hard springs arranged in series necessarily exhibit the same stress applied to their upper part; therefore, they are used only as spacers to make up the system. On the other hand, hard springs located in parallel provide practical resistance to external loads, and thus they quantify the mechanical contribution of the hard domain. Therefore, continuously monitoring the effective volume fraction of the parallel hard spring ($\Phi_{H,P}$) explicitly accounts for the load-transfer capability of the PU system. The total volume fractions of soft (Φ_s) and hard ($\Phi_{H,P} + \Phi_{H,s}$) segments were estimated based on Voronoi tessellation (right of Figure 5a). The effective properties of each spring were defined by fitting the tensile results of the homogeneous domain (Figure 1b) to a third-order Ogden model.

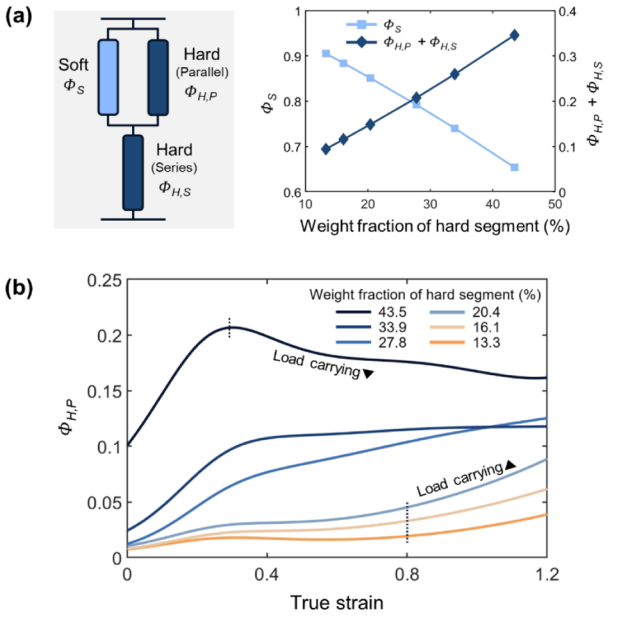


Figure 5. Evaluation of the hard domain load-transfer capability according to the microphase structure. (a) Schematic of the mechanical model considered (left) and changes in the volume fraction of individual phases (right). (b) Change in the effective volume fraction of parallel-connected hard spring during tensile deformation.

The effective volume fraction of the parallel hard springs exhibits distinct trends depending on the microphase structure (Figure 5b). In the globular phase, $\Phi_{H,P}$ is initially close to zero. Because the soft domain effectively dissipated the applied external strain energy, the mechanical contribution of the hard domain was negligible. The $\Phi_{H,P}$ remains low with continuous strain and increases slightly only at $\varepsilon_T > 0.8$ due to chain extension. In contrast, the $\Phi_{H,P}$ of the elongated and bicontinuous phases exhibits substantial values from the onset of the deformation. This is particularly noticeable in a bicontinuous phase in which hard segments are infinitely connected. Importantly, the $\Phi_{H,P}$ of elongated and bicontinuous PUs, which increase linearly with applied strain, exhibit a clear inflection point at $\varepsilon_T > 0.3$. In other words, the load-transfer capability of the hard segment is impaired by continuous deformation.

The strain tensor analysis (Figure 4), clearly demonstrates that when the hard domains are highly connected, instantaneous load transfer is achieved through physical interlocking. Therefore, the decrease in $\Phi_{H,P}$ in the elongated and bicontinuous phases may be related to the fragmentation and dissociation of interconnected hard domains. To make this point clear, an order parameter ($\Delta\psi$) defined by the following equation was adopted:[50,51]

$$\Delta\psi = \frac{1}{N}\sum_{i=1}^{N}\frac{(N_{HS} - N_{SS})2}{(N_{HS} + N_{SS})2} \tag{4}$$

where N_{HS} and N_{SS} indicate the number of hard and soft segments within a 0.75 nm distance from the reference bead *i*. The order parameter quantifies the segment type surrounding the bead *i*. Higher-order parameters indicate that the bead is surrounded more by hard segments and vice versa by soft chains. Therefore, the parameter can be applied to explicitly describe the connectivity between hard domains.

The $\Delta\psi$ of the equilibrated model increased with an increasing hard segment content in the PU molecule (left of Figure 6a). This is reasonable, considering that an increased hard segment content leads to the formation of hard spheres with increased diameters. As the surface-to-volume ratio

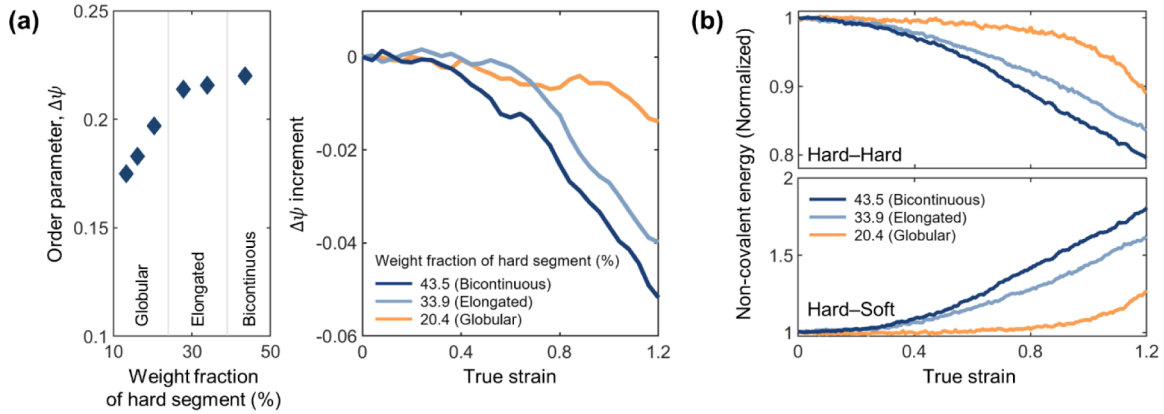


Figure 6. Dissociation of the hard phase due to mechanical tension. (a) Order parameters of the hard segment at the equilibrium state (left) and their increment during stretching (right). (b) Normalized noncovalent energy between PU segments.

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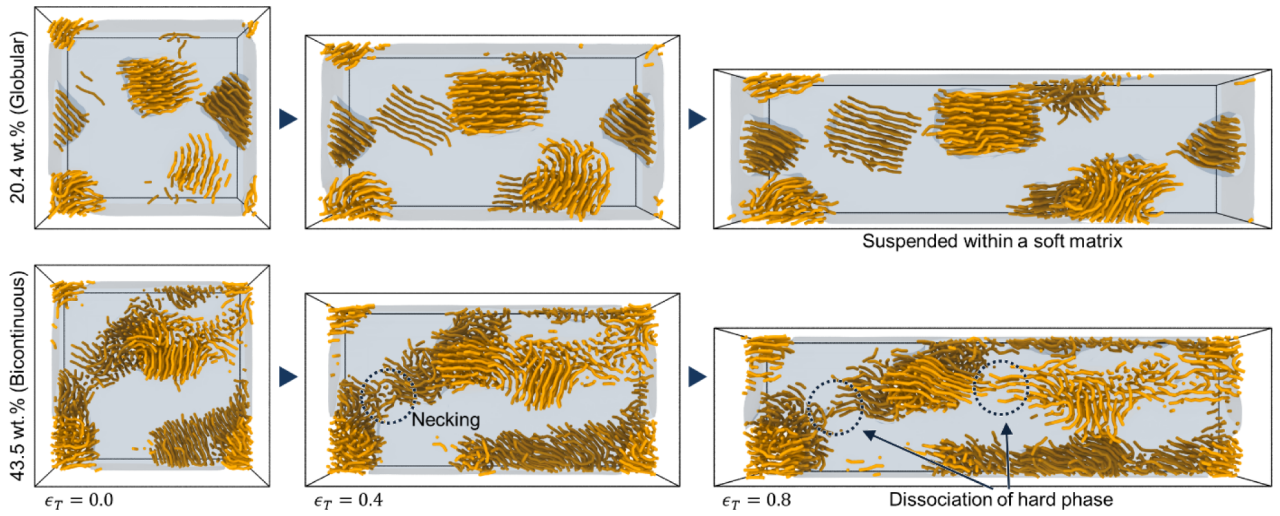


Figure 7. Visualization of the microstructural change in hard domains during the uniaxial deformation. Soft segments are indicated by dark blue transparent blocks.

decreases, fewer hard beads interact with the soft segments. $\Delta\psi$ gradually becomes saturated as the hard spheres connect to each other. When analyzing the increment of $\Delta\psi$ during tensile deformation, clear differences for the microphases can be observed (right of Figure 6a). Elongated and bicontinuous PUs exhibit a sharp decrease in $\Delta\psi$ as the deformation progresses. In other words, the hard phase fragments and dissociates during tension, becoming further surrounded by soft segments. These observations were further supported by the interaction energy analysis (Figure 6b). Although the interaction energy between the hard segments weakens as dissociation progresses, the interaction energy between the hard and soft segments increases. This is interpreted as a highly mobile soft segment occupying the cut surface of the hard domain. In contrast, changes in $\Delta\psi$ and the noncovalent energy are insignificant in globular PU. In other words, the initial bonding relationship of globular hard segments is largely preserved throughout the tensile process.

It is worth noting that the sequential microstructural evolution characteristics of the interconnected hard segments are very similar to those observed for bicontinuous polyureas in the study by Zheng et al.[10] The hard segments of bicontinuous polyurea maintain a highly interconnected structure during initial deformation. However, as the applied strain gradually increases, the hard segments exhibit a global orientation along the tensile direction, providing self-reinforcement to improve the mechanical behavior of the material. The interconnected structure of the hard segments also becomes partially fragmented from this point on. Similarly, the increase in the initial load-carrying efficiency of bicontinuous PU in this study is also highly related to self-reinforcement due to the orientation of the hard segments. The hydrogen bonding between the oriented urethane groups contributes to improved load transfer through the hard domains. On the other hand, the fragmentation of the hard domains, which is dominant at high strains, makes it difficult to transfer the load through the hard segments, which subsequently reduces the load transfer efficiency.

These observations were further supported by microstructural changes (Figure 7). In the 20.4 wt % PU model, the hard domain exhibits orientation and translation behavior along the loading direction. However, the relative distance between the hard segments remains consistent throughout deformation, preventing fragmentation of the hard phase. Conversely, the interlocked structure of the 43.3 wt % model homogeneously distributes the internal strain, prompting significant intermolecular rearrangements of the hard segments. The hard domain undergoes partial necking and dissociation; therefore, the load-transfer capability of the hard domain decreases with phase fragmentation.

The mechanical versatility arising from phase-separated morphology is key to expanding the engineering application of PU. In this regard, understanding the microphase-dependent load transfer capability of the hard segment is important for quantitatively bridging the microstructure−property relationship. This study identifies the underlying mechanisms by which changes in the nanoscale morphology determine macroscopic mechanical properties. This serves as a theoretical framework for rational PU design by promoting a molecular-level understanding of the deformation behavior of individual segments relative to the microphase.

4. CONCLUSION

In this study, we investigated how the microstructure of PU, influenced by the content of hard segments, determines the macroscopic mechanical properties. The load-carrying capacity of the hard segments was carefully analyzed based on the microphase morphology and applied strain.

Increasing the hard segment content in PU molecules leads to a progressive phase evolution from globular to discrete and elongated to bicontinuous domains. Accordingly, the mechanical properties of PU undergo a dramatic transition from hyperelastic to elasto-plastic. The stepwise enhancement of the mechanical properties was largely related to the load-transfer capability of the hard domain in each microphase. In the globular microphase, the strain energy applied during stretching was mostly buffered by the surrounding soft matrix. Therefore, the hard phase underwent little internal deformation, resulting in minimal intervention of the mechanical properties. In elongated and bicontinuous PUs, the strain that evolved during tension was distributed homogeneously in each phase due to the interlocked structure. Therefore, an effective load transfer to the hard domain occurred immediately after the onset of the deformation.

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The load-transfer capability of the hard phase, depending on the microscopic structure, was further analyzed by using a mechanical model. A model in which one soft hyperelastic spring was coupled with two hard elasto-plastic springs was considered, and the effective weight of the parallel-located hard spring was evaluated during deformation. For globular PU, the load-transfer efficiency of the hard domain increased with the applied strain because load transfer through the polymer backbone chain was promoted by the expansion of the PU molecules at large deformation. In contrast, in the elongated and bicontinuous phases, the hard domains underwent fragmentation and dissociation with continuous deformation. Therefore, the load-transfer efficiency was highest in the initial stage of deformation and decreased with the applied strain. Consequently, this study provides a molecular-level understanding of the deformation behavior and macroscopic mechanical responses of microphase-separated PU.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.4c01773.

Covalent and noncovalent parameters for consisting the PU chains in the MARTINI force field (Tables S1 and S2); configurations of microphase-separated PU microstructures (Figure S1); spring connection model to represent the heterogeneous structure of PU (Figures S2 and S3) (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Das, A.; Mahanwar, P. A brief discussion on advances in polyurethane applications. *Adv. Ind. Eng. Polym. Res.* **2020**, *3* (3), 93−101.

(2) Yu, W.; Du, M.; Zhang, D.; Lin, Y.; Zheng, Q. Influence of dangling chains on molecular dynamics of polyurethanes. *Macromolecules* **2013**, *46* (18), 7341−7351.

(3) Ma, R.; Wang, Y.; Huang, H.; Zhao, X.; Li, X.; Zhang, L.; Gao, Y. Insight into the Fracture Properties and Molecular Mechanism of Polyurethane. *Macromolecules* **2024**, *57* (9), 4484−4494.

(4) Velankar, S.; Cooper, S. L. Microphase separation and rheological properties of polyurethane melts. 2. Effect of block incompatibility on the microstructure. *Macromolecules* **2000**, *33* (2), 382−394.

(5) Velankar, S.; Cooper, S. L. Microphase separation and rheological properties of polyurethane melts. 3. Effect of block incompatibility on the viscoelastic properties. *Macromolecules* **2000**, *33* (2), 395−403.

(6) Cheng, B. X.; Gao, W. C.; Ren, X. M.; Ouyang, X. Y.; Zhao, Y.; Zhao, H.; Wu, W.; Huang, C. X.; Liu, Y.; Liu, X. Y.; Li, H. N.; Li, R. K. A review of microphase separation of polyurethane: Characterization and applications. *Polym. Test.* **2022**, *107*, 107489.

(7) Garrett, J. T.; Runt, J.; Lin, J. S. Microphase separation of segmented poly (urethane urea) block copolymers. *Macromolecules* **2000**, *33* (17), 6353−6359.

(8) Zhang, R.; Yu, S.; Chen, S.; Wu, Q.; Chen, T.; Sun, P.; Li, B.; Ding, D. Reversible cross-linking, microdomain structure, and heterogeneous dynamics in thermally reversible cross-linked polyurethane as revealed by solid-state NMR. *J. Phys. Chem. B* **2014**, *118* (4), 1126−1137.

(9) Mokeev, M. V.; Ostanin, S. A.; Saprykina, N. N.; Zuev, V. V. Microphase structure of polyurethane-polyurea copolymers as revealed by solid-state NMR: Effect of molecular architecture. *Polymer* **2018**, *150*, 72−83.

(10) Zheng, T.; Li, T.; Shi, J.; Wu, T.; Zhuang, Z.; Xu, J.; Guo, B. Molecular insight into the toughness of polyureas: A hybrid all-atom/coarse-grained molecular dynamics study. *Macromolecules* **2022**, *55* (8), 3020−3029.

(11) Klinedinst, D. B.; Yilgör, I.; Yilgör, E.; Zhang, M.; Wilkes, G. L. The effect of varying soft and hard segment length on the structure−property relationships of segmented polyurethanes based on a linear symmetric diisocyanate, 1, 4-butanediol and PTMO soft segments. *Polymer* **2012**, *53* (23), 5358−5366.

(12) Wang, Y.; Ma, R.; Li, H.; Hu, S.; Gao, Y.; Liu, L.; Zhao, X.; Zhang, L. Effect of the content and strength of hard segment on the viscoelasticity of the polyurethane elastomer: Insights from molecular dynamics simulation. *Soft Matter* **2022**, *18* (21), 4090−4101.

(13) Rahmawati, R.; Nozaki, S.; Kojio, K.; Takahara, A.; Shinohara, N.; Yamasaki, S. Microphase-separated structure and mechanical properties of cycloaliphatic diisocyanate-based thiourethane elastomers. *Polym. J.* **2019**, *51* (2), 265−273.

(14) Furukawa, M.; Mitsui, Y.; Fukumaru, T.; Kojio, K. Microphase-separated structure and mechanical properties of novel polyurethane elastomers prepared with ether based diisocyanate. *Polymer* **2005**, *46* (24), 10817−10822.

(15) Merino, D. H.; Feula, A.; Melia, K.; Slark, A. T.; Giannakopoulos, I.; Siviour, C. R.; Buckley, C. P.; Greenland, B. W.; Liu, D.; Gan, Y.; Harris, P. J.; Chippindale, A. M.; Hamley, I. W.; Hayes, W. A systematic study of the effect of the hard end-group composition on the microphase separation, thermal and mechanical properties of supramolecular polyurethanes. *Polymer* **2016**, *107*, 368−378.

(16) Tien, Y. I.; Wei, K. H. Hydrogen bonding and mechanical properties in segmented montmorillonite/polyurethane nanocomposites of different hard segment ratios. *Polymer* **2001**, *42* (7), 3213−3221.

(17) Chen, C.; Ying, W. B.; Li, J.; Kong, Z.; Li, F.; Hu, H.; Tian, Y.; Kim, D. H.; Zhang, R.; Zhu, J. A self-healing and ionic liquid affiliative polyurethane toward a piezo 2 protein inspired ionic skin. *Adv. Funct. Mater.* **2022**, *32* (4), 2106341.

(18) Zeng, Y.; Chen, Y.; Sha, D.; Wu, Y.; Qiu, R.; Liu, W. Highly stretchable fatty acid chain-dangled thermoplastic polyurethane elastomers enabled by H-bonds and molecular chain entanglements. *ACS Sustainable Chem. Eng.* **2022**, *10* (35), 11524−11532.

(19) Qi, H. J.; Boyce, M. C. Stress−strain behavior of thermoplastic polyurethanes. *Mech. Mater.* **2005**, *37* (8), 817−839.

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(20) Petrović, Z. S.; Ferguson, J. Polyurethane elastomers. *Prog. Polym. Sci.* **1991**, *16* (5), 695−836.
(21) Estes, G. M.; Seymour, R. W.; Cooper, S. L. Infrared studies of segmented polyurethane elastomers. II. Infrared dichroism. *Macromolecules* **1971**, *4* (4), 452−457.
(22) Huang, M.; Li, Z.; Xu, W.; He, S.; Liu, W.; Jiang, L.; Liu, H. Visualization and quantification of microphase separation in thermoplastic polyurethanes under different hard segment contents and its effect on the mechanical properties. *Polym. Test.* **2024**, *131*, 108329.
(23) Akindoyo, J. O.; Beg, M.; Ghazali, S.; Islam, M. R.; Jeyaratnam, N.; Yuvaraj, A. R. Polyurethane types, synthesis and applications−a review. *RSC Adv.* **2016**, *6* (115), 114453−114482.
(24) Somarathna, H. M. C. C.; Raman, S. N.; Mohotti, D.; Mutalib, A. A.; Badri, K. H. The use of polyurethane for structural and infrastructural engineering applications: A state-of-the-art review. *Constr. Build. Mater.* **2018**, *190*, 995−1014.
(25) Joseph, J.; Patel, R. M.; Wenham, A.; Smith, J. R. Biomedical applications of polyurethane materials and coatings. *Trans. IMF* **2018**, *96* (3), 121−129.
(26) Naureen, B.; Haseeb, A. S. M. A.; Basirun, W. J.; Muhamad, F. Recent advances in tissue engineering scaffolds based on polyurethane and modified polyurethane. *Mater. Sci. Eng.* **2021**, *118*, 111228.
(27) Vervaeke, R.; Debruyne, S.; Vandepitte, D. Numerical and experimental analysis of vibration damping performance of polyurethane adhesive in machine operations. *Int. J. Adhes. Adhes.* **2019**, *90*, 47−54.
(28) Verdejo, R.; Stämpfli, R.; Alvarez-Lainez, M.; Mourad, S.; Rodriguez-Perez, M. A.; Brühwiler, P. A.; Shaffer, M. Enhanced acoustic damping in flexible polyurethane foams filled with carbon nanotubes. *Compos. Sci. Technol.* **2009**, *69* (10), 1564−1569.
(29) Thompson, A. P.; Aktulga, H. M.; Berger, R.; Bolintineanu, D. S.; Brown, W. M.; Crozier, P. S.; in't Veld, P. J.; Kohlmeyer, A.; Moore, S. G.; Nguyen, T. D.; et al. LAMMPS-a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Comput. Phys. Commun.* **2022**, *271*, 108171.
(30) Kwon, M. S.; Yu, C.; Ha, J.-K.; Ahn, J.; Lee, J.; Choi, J.; Chang, T.; Min, S. K. Defect-free acrylic polymers with a near-Poisson distribution prepared via catalyst-free visible-light-driven radical polymerization. *ChemRxiv* **2023**.
(31) Marrink, S. J.; Risselada, H. J.; Yefimov, S.; Tieleman, D. P.; De Vries, A. H. The MARTINI force field: Coarse grained model for biomolecular simulations. *J. Phys. Chem. B* **2007**, *111* (27), 7812−7824.
(32) Kim, H.; Won, D.; Ko, S. H.; Choi, J. Thermodynamic Mechanism Governing the Coalescence of Conductive Particles in PEDOT: PSS under Laser Irradiation. *Macromolecules* **2024**, *57* (5), 2048−2056.
(33) Ghermezcheshme, H.; Makki, H.; Mohseni, M.; Ebrahimi, M.; de with, G. MARTINI-based simulation method for step-growth polymerization and its analysis by size exclusion characterization: A case study of cross-linked polyurethane. *Phys. Chem. Chem. Phys.* **2019**, *21* (38), 21603−21614.
(34) Kizilkaya, D.; Ghermezcheshme, H.; Sabzevar, S. E.; Makki, H.; Kacar, G. Modeling surface segregation of smart PU coatings at hydrophilic and hydrophobic interfaces via coarse-grained molecular dynamics and mesoscopic simulations. *Prog. Org. Coat.* **2023**, *174*, 107279.
(35) Ghermezcheshme, H.; Makki, H.; Mohseni, M.; Ebrahimi, M. Hydrophilic dangling chain interfacial segregation in polyurethane networks at aqueous interfaces and its underlying mechanisms: Molecular dynamics simulations. *Phys. Chem. Chem. Phys.* **2020**, *22* (45), 26351−26363.
(36) Vakili, H.; Mohseni, M.; Makki, H.; Yahyaei, H.; Ghanbari, H.; González, A.; Irusta, L. Self-assembly of a patterned hydrophobic-hydrophilic surface by soft segment microphase separation in a segmented polyurethane: Combined experimental study and molecular dynamics simulation. *Polymer* **2020**, *195*, 122424.
(37) Heyes, D. M. Pressure tensor of partial-charge and point-dipole lattices with bulk and surface geometries. *Phys. Rev. B* **1994**, *49* (2), 755.
(38) Rohm, K.; Manas-Zloczower, I. A micromechanical approach to TPU mechanical properties: Framework and experimental validation. *Mech. Mater.* **2023**, *180*, 104627.
(39) Sui, T.; Salvati, E.; Ying, S.; Sun, G.; Dolbnya, I. P.; Dragnevski, K.; Prisacariu, C.; Korsunsky, A. M. Strain softening of nano-scale fuzzy interfaces causes Mullins effect in thermoplastic polyurethane. *Sci. Rep.* **2017**, *7* (1), 916.
(40) Stanford, J. L.; Young, R. J.; Day, R. J. Formation and properties of urethane-diacetylene segmented block copolymers. *Polymer* **1991**, *32* (9), 1713−1725.
(41) Yang, S. J.; Rosenbloom, S. I.; Fors, B. P.; Silberstein, M. N. Elucidating the impact of microstructure on mechanical properties of phase-segregated polyurea: Finite element modeling of molecular dynamics derived microstructures. *Mech. Mater.* **2024**, *188*, 104863.
(42) Kim, H.; Choi, J. Subcontinuum interpretation of mechanical behavior for cross-linked epoxy networks. *Macromolecules* **2022**, *55* (14), 5916−5925.
(43) Kim, H.; Choi, J. Interfacial and mechanical properties of liquid crystalline elastomer nanocomposites with grafted Au nanoparticles: A molecular dynamics study. *Polymer* **2021**, *218*, 123525.
(44) Rycroft, C. H. VORO++: A three-dimensional Voronoi cell library in C++. *Chaos* **2009**, *19*, 041111.
(45) Kim, H.; Kim, H.; Choi, J. Interface mechanics of liquid crystal polymer nanocomposites with high concentrations of MWCNTs. *Compos. Sci. Technol.* **2022**, *222*, 109376.
(46) Kim, Y.; Kim, H.; Choi, J. Mechanical origin of shape memory performance for crosslinked epoxy networks. *Eur. Polym. J.* **2023**, *194*, 112162.
(47) Kim, H.; Choi, J. Mechanical assessment of interfacial stability of LCP/MWCNT nanocomposites during phase transition. *Compos., Part A* **2023**, *167*, 107461.
(48) Kim, H.; Choi, J. Theoretical analysis of phase transition behavior of ALCP/CNT nanocomposites interface by photo and thermal stimulation. *Compos., Part A* **2023**, *175*, 107824.
(49) Kim, H.; Choi, J. Molecular-scale mechanics of a crystalline polyethylene interface formed by projection of carbon nanotube structures. *Surf. Interfaces* **2024**, *47*, 104213.
(50) Rex, M.; Löwen, H. Influence of hydrodynamic interactions on lane formation in oppositely charged driven colloids. *Eur. Phys. J. E* **2008**, *26*, 143−150.
(51) Dutta, S.; Chakrabarti, J. Transient dynamical responses of a charged binary colloid in an electric field. *Soft Matter* **2018**, *14* (22), 4477−4482.

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